

Project ID:

1. Topic (12 words max)

“PaddyGuard AI: Multimodal Intelligent System for Rice Disease and Pest Management”

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Rice farmers - particularly smallholders with limited agronomic support—suffer large yield losses from leaf diseases (e.g., bacterial blight, leaf blast, brown spot) and insect pests. Timely, accurate diagnosis is constrained by limited extension services, low literacy, and irregular access to laboratory testing, causing delayed or incorrect treatments that reduce yields and incomes and increase pesticide misuse. This project addresses the need for an accessible, field-ready decision support system that combines image and voice inputs, reliable out-of-distribution (OOD) safeguards, and a knowledge-driven advisory to reduce misdiagnosis and unsafe chemical use.

Socially and economically, improved diagnosis and calibrated treatment recommendations increase food security and farmer income while protecting labor and consumer health. Technically, the problem demands robust multimodal AI (speech → symptom extraction, image-based classification, OOD detection, RAG retrieval) and explainability to be trusted in-field. Environmentally and ethically, reducing unnecessary pesticide application mitigates ecosystem damage and human exposure; the dosage calculator and knowledge-based rules aim to prevent misuse and support responsible chemical application. Legal/regulatory concerns (pesticide compliance, data privacy for farmer-supplied data) are considered in design and deployment. The project sits at the intersection of AI/ML, Computer Vision, and Agricultural Informatics (FoC research cluster) and targets the AgriTech industry vertical (precision diagnostics, advisory systems).

Contribution to SDGs: it directly supports SDG 2 (Zero Hunger) by improving crop health and yields, SDG 3 (Good Health) by reducing harmful chemical exposure, and SDG 12 (Responsible Consumption and Production) via optimized pesticide use. The system’s focus on low-literacy voice interfaces increases inclusiveness for marginalized farmers, aligning technical innovation with social impact.

References

- [1] S. Sladojevic, M. Arsenovic, A. Anderla, D. Culibrk and D. Stefanovic, "Deep neural networks based recognition of plant diseases," *Computational Intelligence and Neuroscience*, vol. 2016, Article ID 3289801, 2016.
- [2] J. Amara, B. Bouaziz and A. Algergawy, "A deep learning-based approach for banana leaf diseases classification," *Proc. ICIAR Workshops*, 2017.
- [3] O. Ronneberger, P. Fischer and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. MICCAI*, 2015, pp. 234–241.
- [4] J. Devlin, M.-W. Chang, K. Lee and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

6. Brief description of Existing Research and Systems (300 words max)

Automated crop disease diagnosis systems have emerged over the past decade, often leveraging **computer vision and deep learning** to identify plant stress from image data. A prominent commercial example is **Plantix**, a mobile app that uses deep learning to analyze photos of crops for pests, diseases, and nutrient deficiencies and provides management recommendations, though its efficacy claims and ecological impact have been questioned due to its industry-linked pesticide recommendations [1]. Other mobile and web tools (e.g., Agrio, Plant Disease ID) apply similar image-based detection but suffer from limitations such as poor accuracy, restricted crop coverage, or dependency on internet connectivity [2].

In research, convolutional neural networks (CNNs) remain the dominant paradigm for plant disease detection. Recent studies have explored lightweight architectures such as EfficientNet, MobileNetV2, and ShuffleNet for rice leaf disease identification optimized for mobile deployment, achieving high accuracy under controlled conditions [3]. Hybrid models combining attention mechanisms or ensemble CNNs have shown further performance improvements (with accuracies above 98 %) in lab settings [4], and resource-optimized models have been deployed on microcontrollers for real-time detection on edge devices, mitigating connectivity dependency [5]. Reviews of the field note significant progress in CNN-based detection but highlight substantial performance drops when these models are applied to unpredictable field environments, implicating **generalization challenges** and data diversity gaps [6].

Despite high classification accuracies in research settings, existing systems predominantly focus on **image data only**, lack **multimodal inputs** (e.g., speech, sensor data), and seldom incorporate **robust out-of-distribution detection**, explainability, or actionable agricultural advisory integrated with agronomic knowledge. Many tools also **target a limited set of diseases** and neglect pests or early symptom stages. These limitations underscore the need for more comprehensive, robust, and farmer-centric systems—justifying the *PaddyGuard AI* research focus.

References

- [1] Plantix, Wikipedia, 2026. Available: <https://en.wikipedia.org/wiki/Plantix>.
- [2] "Evaluating Plant Disease Detection Mobile Applications: Quality and Limitations," *MDPI Agronomy*, 2025.
- [3] K. Saddami et al., "Efficient CNNs for Rice Leaf Disease Identification," *arXiv*, 2024.
- [4] "Deep learning-based automatic diagnosis of rice leaf diseases using ensemble CNN models," *Scientific Reports*, 2025.

[5] H. Nugroho et al., "Resource-optimized CNNs for real-time rice disease detection," *Plant Methods*, 2024.

[6] "Recent advances in plant disease detection: challenges and opportunities," *Plant Methods*, 2025.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed **PaddyGuard AI** system is a **multimodal, intelligent decision support platform** for rice disease and pest management. It is designed to overcome the key limitations identified in existing systems (Section 6), namely: reliance on image-only inputs, poor robustness in real-world conditions, absence of uncertainty handling, and lack of actionable, knowledge-grounded treatment guidance.

Addressing Identified Gaps

Unlike conventional single-modal solutions, PaddyGuard AI integrates **image-based deep learning**, **voice-based symptom analysis**, **out-of-distribution (OOD) detection**, and a **hybrid AI + knowledge-based advisory system**. OOD detection prevents unreliable predictions when unseen diseases, pests, or non-rice images are provided, improving field reliability [3]. Voice-based diagnosis enables accessibility for **low-literacy farmers**, a major gap in current tools [4]. Furthermore, the system combines predictive AI outputs with **rule-based expert knowledge and Retrieval-Augmented Generation (RAG)** to deliver explainable and safe treatment recommendations rather than raw predictions [5].

System Perspective: Input–Process–Output

Inputs

- Rice leaf images (captured via mobile devices)
- Pest images
- Farmer voice descriptions of symptoms
- Contextual metadata (optional: crop stage, location)

Process

1. **Preprocessing:** Image normalization, augmentation; speech-to-text conversion
2. **AI Inference:**
 - CNN-based disease and pest classification
 - OOD detection to filter unknown/unreliable samples [3]
 - NLP-based symptom classification using transformer models [6]
3. **Decision Layer:**
 - Rule-based validation
 - Dosage calculator

- RAG-based chatbot retrieves agronomic knowledge

4. Explainability & Confidence Calibration

Outputs

- Disease or pest identification with confidence level
- OOD warning for unknown cases
- Treatment recommendations (chemical & non-chemical)
- Dosage guidance
- Voice + text advisory responses

References

- [1] S. Sladojevic et al., “Deep neural networks based recognition of plant diseases,” Computational Intelligence and Neuroscience, 2016.
- [2] J. Amara et al., “A deep learning-based approach for banana leaf diseases classification,” ICIAR, 2017.
- [3] S. Liang, Y. Li, and R. Srikant, “Enhancing the reliability of out-of-distribution image detection,” ICLR, 2018.
- [4] A. Radford et al., “Robust speech recognition via large-scale weak supervision,” arXiv:2212.04356, 2022.
- [5] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” NeurIPS, 2020.
- [6] J. Devlin et al., “BERT: Pre-training of deep bidirectional transformers,” NAACL-HLT, 2019.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful execution of **PaddyGuard AI** requires interdisciplinary expertise spanning **agriculture, artificial intelligence, and software engineering**. The project team will acquire and strengthen the required domain knowledge, technical skills, and datasets through structured academic study, field engagement, and experimental research.

Acquisition of Specialized Domain Expertise

Agricultural domain expertise—specifically in **rice crop diseases, pests, and treatment practices**—will be gained through multiple channels. Team members will study authoritative literature published by agricultural research bodies such as the **International Rice Research Institute (IRRI)** and peer-reviewed journals focusing on rice pathology and pest management [1], [2]. In addition, knowledge will be reinforced through **consultations with agricultural extension officers and local farmers**, enabling the team to understand real-world symptom descriptions, farming practices, and decision-making challenges. Field visits to paddy cultivation sites will further support practical understanding and contextual relevance.

Acquisition of Technical Knowledge and Skills

The technical expertise required for this project includes **deep learning, computer vision, natural language processing, speech recognition, and retrieval-augmented generation (RAG)**. Team members will develop these competencies through undergraduate coursework in machine learning, artificial intelligence, and software engineering, supplemented by self-directed learning using research papers, online documentation, and tutorials. Hands-on experimentation with frameworks such as **PyTorch, TensorFlow, Hugging Face Transformers, and OpenCV** will be central to skill development [3], [4]. Each member will focus on a specific project component, ensuring depth of expertise and clear individual contribution.

Data Collection Strategy

The project will utilize a **combination of publicly available datasets and primary data collection**. Existing rice disease and pest image datasets (e.g., PlantVillage-derived subsets and agricultural research datasets) will be used for baseline model training [5]. To improve robustness and field relevance, the team plans to **collect real images from local paddy fields** under varying lighting and environmental conditions. Symptom description data will be collected through **farmer interviews and voice recordings**, which will be transcribed and anonymized for NLP model training. All collected data will be curated, labeled, and split into training, validation, and testing sets following standard ML practices.

Ethical Clearance Considerations

This project involves **non-invasive data collection** from farmers (images and voice descriptions) and does not involve medical or sensitive personal data. However, ethical considerations such as **informed consent, data anonymization, and privacy protection** will be strictly followed. If required by institutional guidelines, ethical clearance will be sought prior to field data collection to ensure compliance with research ethics standards [6].

References

- [1] IRRI, "Rice Knowledge Bank," International Rice Research Institute, 2024.
- [2] S. Sladojevic et al., "Deep neural networks based recognition of plant diseases," *Computational Intelligence and Neuroscience*, 2016.
- [3] J. Devlin et al., "BERT: Pre-training of deep bidirectional transformers," *NAACL-HLT*, 2019.
- [4] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," *NeurIPS*, 2020.
- [5] D. P. Hughes and M. Salathé, "An open access repository of images on plant health," *arXiv:1511.08060*, 2015.
- [6] World Medical Association, "Declaration of Helsinki – Ethical Principles for Research," 2013.

9. Objectives and Novelty

Main Objective

To design and develop **PaddyGuard AI**, a robust and farmer-centric **multimodal intelligent decision support system** that accurately detects rice leaf diseases and pests, handles unseen field conditions using out-of-distribution detection, and provides safe, explainable, and knowledge-driven treatment recommendations through image and voice-based interaction.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22273680	Voice-based symptom understanding and disease classification.	- Collect voice-based symptom descriptions from farmers and agricultural sources - Convert speech to text using automatic speech recognition models - Clean and normalize symptom text (stop-word removal, stemming, domain term mapping) - Train and compare traditional ML (TF-IDF + SVM) and transformer-based NLP models - Evaluate classification accuracy and robustness	- Enables voice-first disease diagnosis for farmers with limited literacy - Uses natural language symptom descriptions, not structured inputs - Comparative evaluation of traditional vs transformer-based NLP for agricultural diagnosis

		Convert diagnosis output to synthesized speech for low-literacy users	
	Image-based rice disease detection with reliability enhancement.	- Prepare and preprocess rice leaf image datasets - Train CNN models using transfer learning (ResNet, EfficientNet, etc.) - Evaluate model performance under controlled and real-world conditions - Implement OOD detection to identify unseen or non-rice images - Calibrate confidence thresholds and analyze false prediction reduction	- Integrates OOD detection into rice disease classification, rarely addressed in prior work - Prevents over-confident wrong predictions in real field conditions - Improves trustworthiness of AI decisions for practical agricultural deployment
	Treatment advisory system and agricultural chatbot.	- Design rule-based expert system for treatment validation - Develop pesticide dosage calculator based on crop stage and disease - Build agricultural knowledge base using curated documents - Implement RAG-based chatbot to retrieve accurate agronomic knowledge	- Combines AI predictions with expert rules, reducing unsafe recommendations - Introduces dosage-aware treatment guidance, not just disease identification - RAG-based advisory improves explainability and farmer decision confidence

		Integrate diagnosis outputs with advisory system	
	Image-based pest identification and unknown pest detection.	- Collect and prepare rice pest image datasets - Train pest classification or detection models - Evaluate performance under varying backgrounds - Implement OOD detection for unseen pests - Integrate pest results into overall decision support flow	- Extends beyond disease detection to integrated pest analysis - Detects unknown pests instead of forcing incorrect classifications - Supports holistic crop health management

**10. This part is to be filled by the
Supervisor and the Co-supervisor of the Project.**

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	☒	No	☐
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- b) The proposed topic exhibits novelty.

Yes	☒	No	☐
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- c) The student group can successfully execute the proposed project.

Yes	☒	No	☐
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- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	☒	No	☐
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- e) Any other comments:

Find the datasets

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Bhagya Nathali	Silva	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				